ConfigStream
Evidence-oriented configuration history

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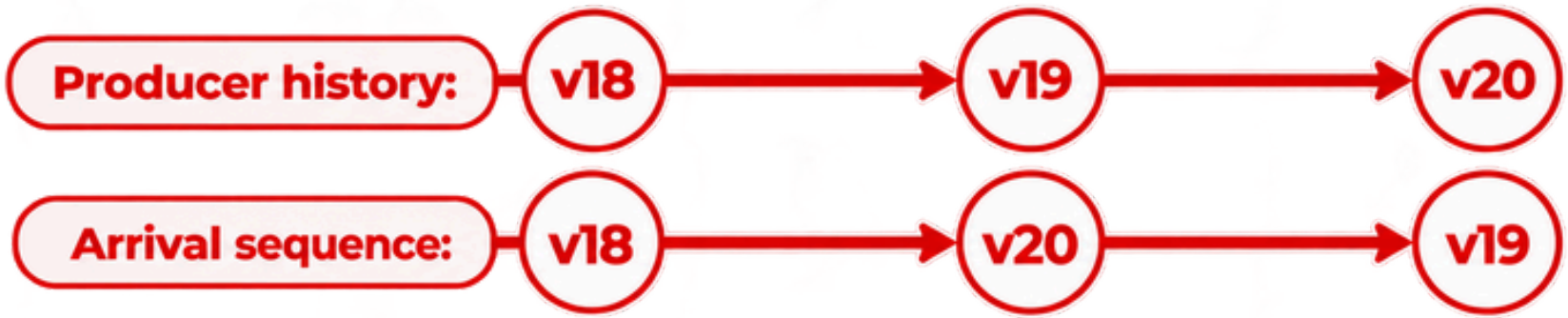
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01

INTRODUCTION

Problem statement



A late record can be valid.

Arrival order cannot define history.

A wrong pair can create a false diff or alert.

The previous arrival is not necessarily the previous version

Why this matter

late arrival → wrong pairing → false change → difficult audit



missing predecessor

duplicate record

projection updated before history commit.

02

RESEARCH QUESTION

Research question

1. Can the system identify valid adjacent states under disorder, duplicates, bounded state, and restart-sensitive execution?

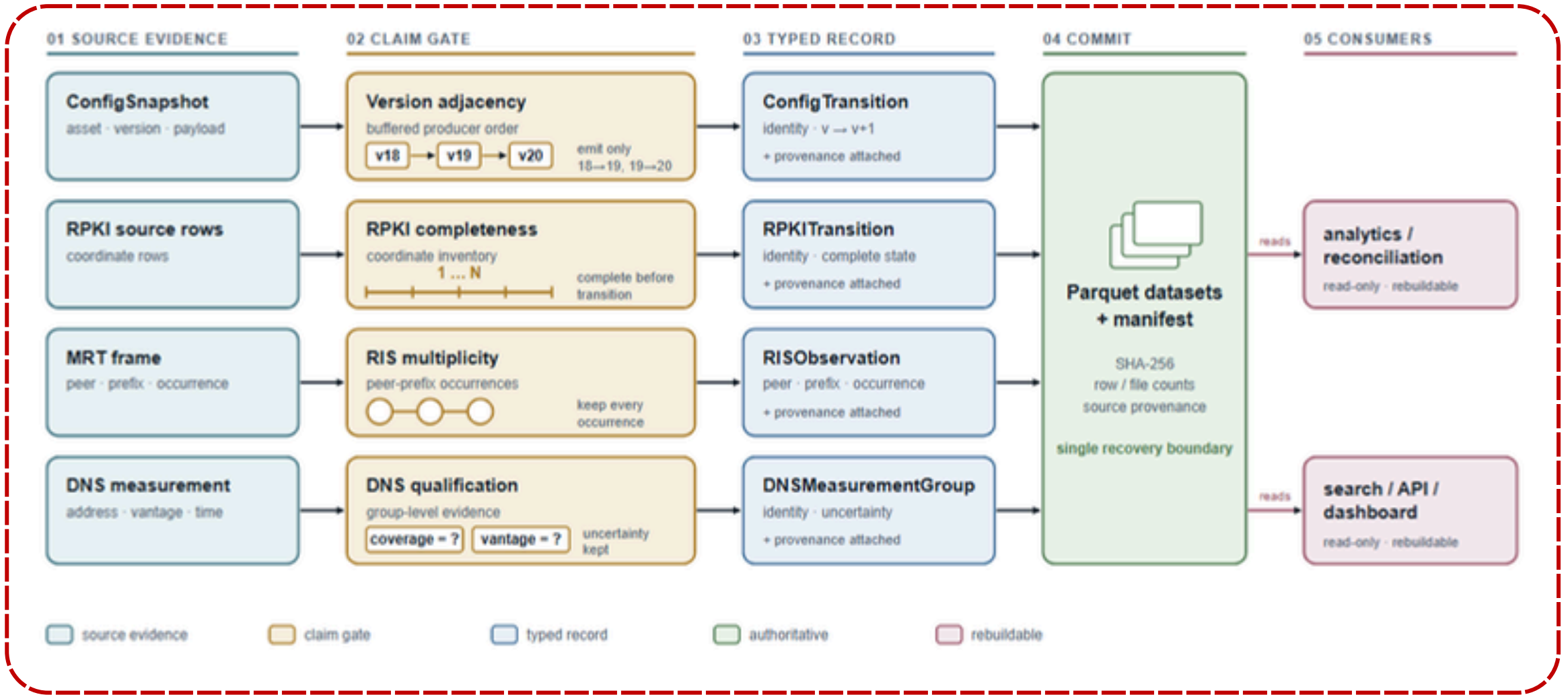
2. Can committed history remain independent from rebuildable search and API views?

3. How should correctness and performance be compared when workloads use different evidence units?

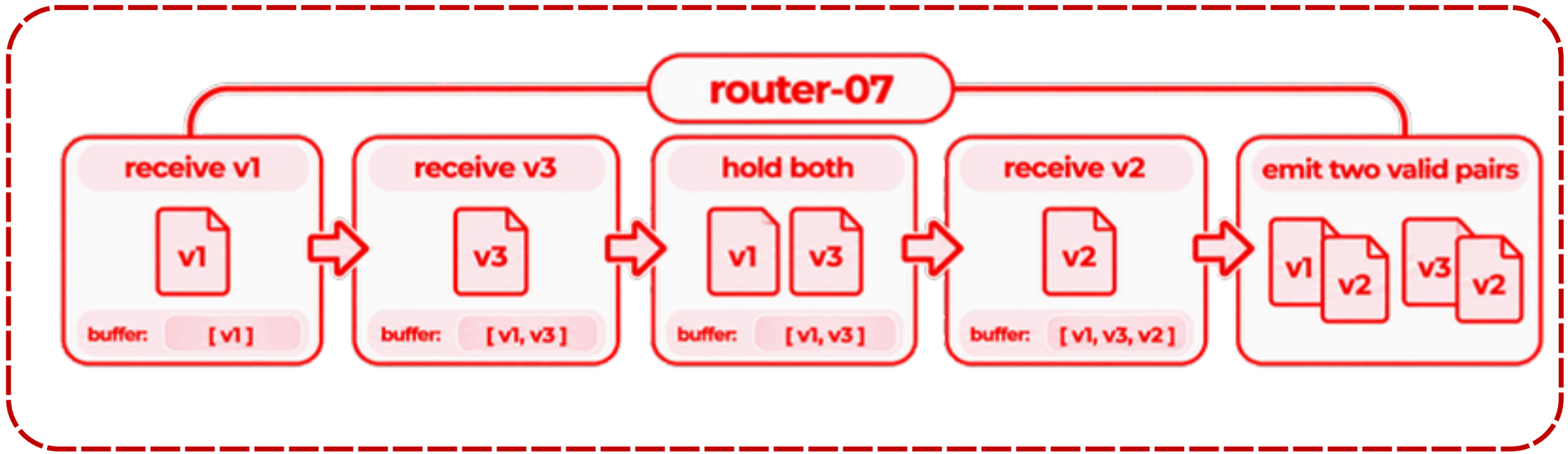
03

METHODOLOGY

Method Overview

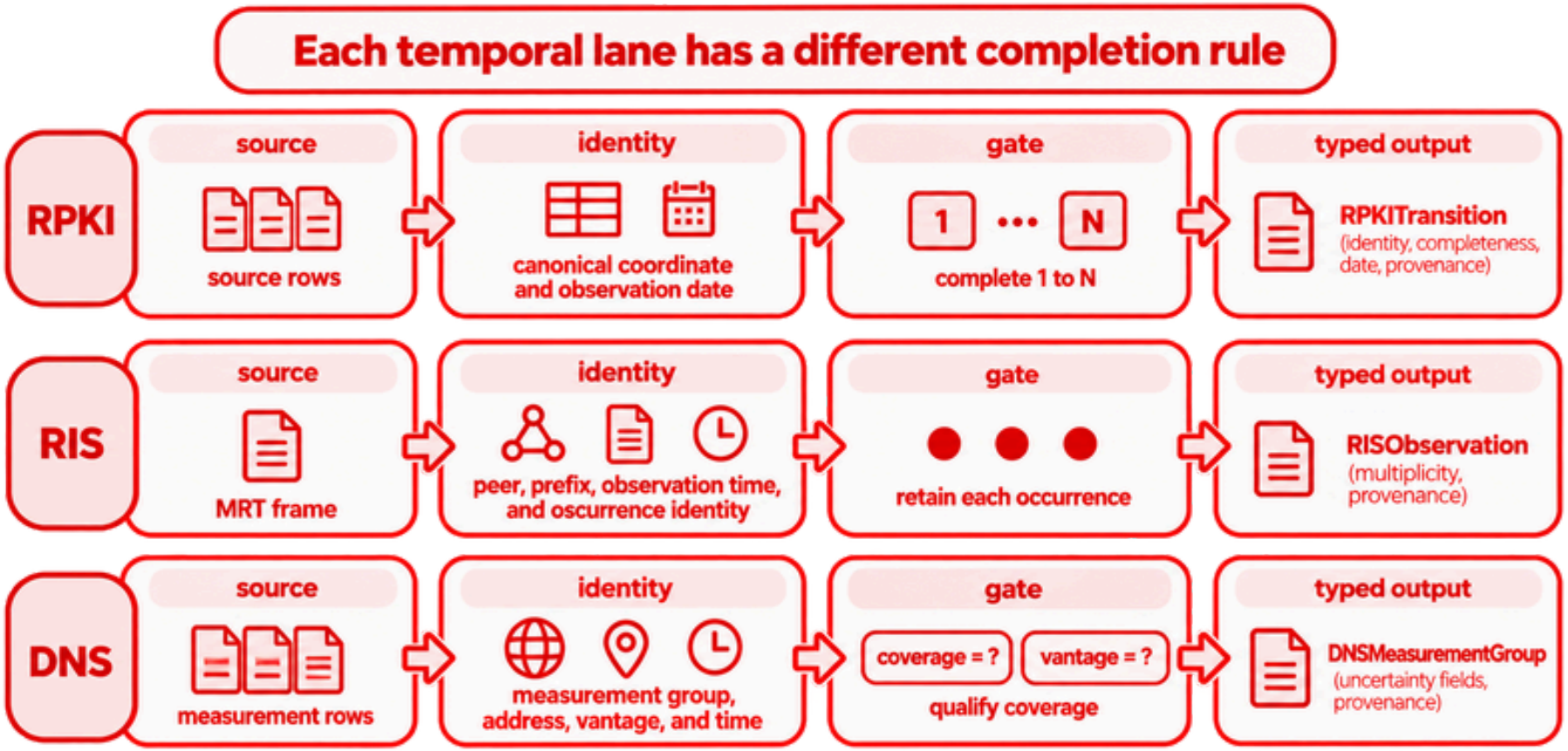


The buffer waits for the missing predecessor

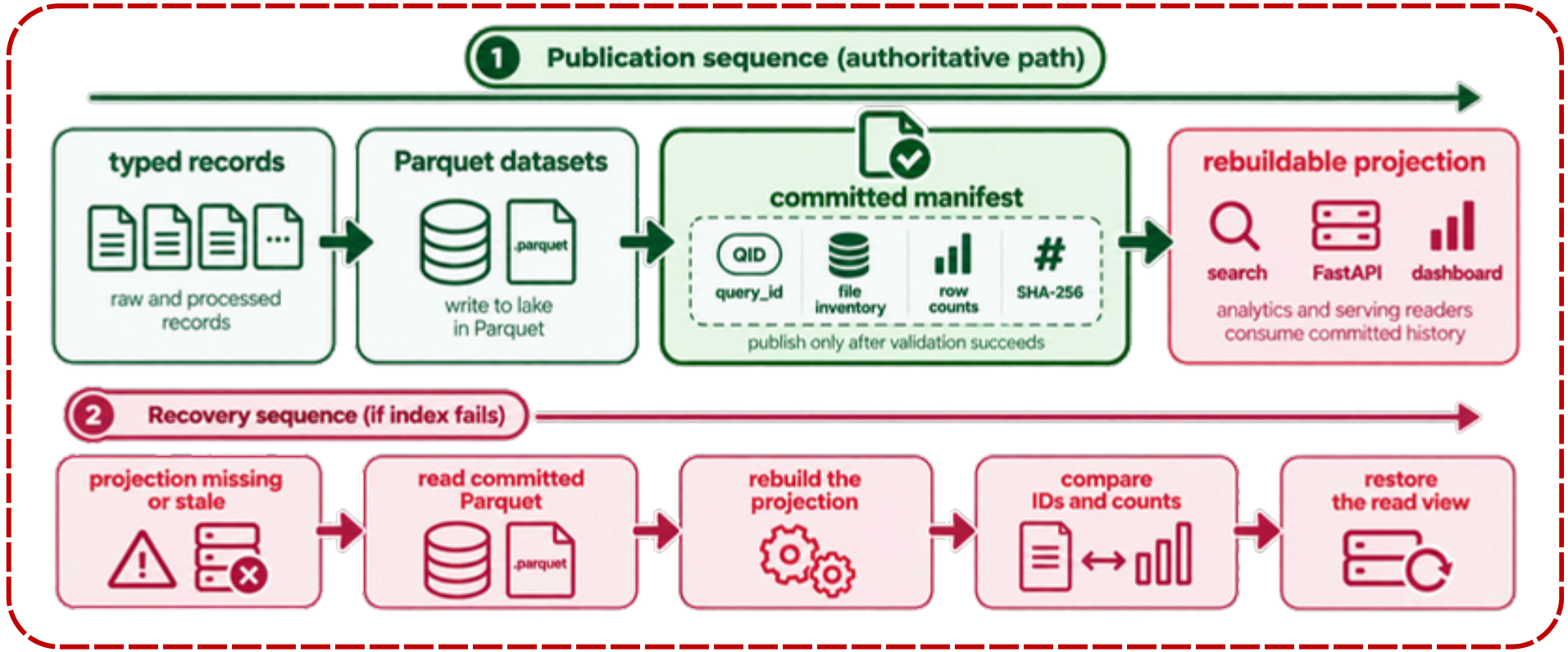


- Bind every snapshot to an “asset_id” and a producer version before comparing it.
- Deduplicate by record identity before emission.
- Sort retained records by producer version, not by arrival time.
- Emit only when two records belong to the same asset and the successor version is exactly one greater.
- Keep unresolved gaps, conflicts, late records, and evictions as diagnostics.

Workload-specific temporal evidence



Publication and recovery



04

EXPERIMENT SETUP

Design

Workload	Unit	Accepted role
TerraDS	File version	End-to-end configuration-history replay
RPKI	Canonical row or member	Row-oriented processing and storage
RIS	Peer-prefix occurrence	Multiplicity-preserving temporal processing
DNS	Measurement or grouped row	Grouped temporal comparison

Result: TerraDS replay

10,188

snapshots

3,683

adjacent pairs

68.742s

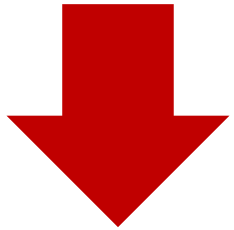
completion
including drain

Result: RPKI storage layouts

R1: 47,756,423 bytes

R2: 3,555,516 bytes

R3: 6,954,890 bytes



92.55% reduction for R2

85.44% reduction for R3

05

DISCUSSION

Result: RPKI storage layouts

- **RIS replay: Peer-prefix multiplicity and run-boundary completeness remain unverified.**
- **End-to-end recovery: Recovery from committed Parquet under interrupted publication and repeated faults remains untested.**
- **Operational scalability: Throughput, latency, and memory at multi-node scale or larger state spaces remain unevaluated.**

NEXT STEP

1. Freeze the evidence pack

2. Run the paced RIS replay

3. Execute projection and checkpoint fault tests

THANK YOU